Vision System for Yarn Error Detection on a Custom Knitting Machine

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Motivation

Undetected defects issues

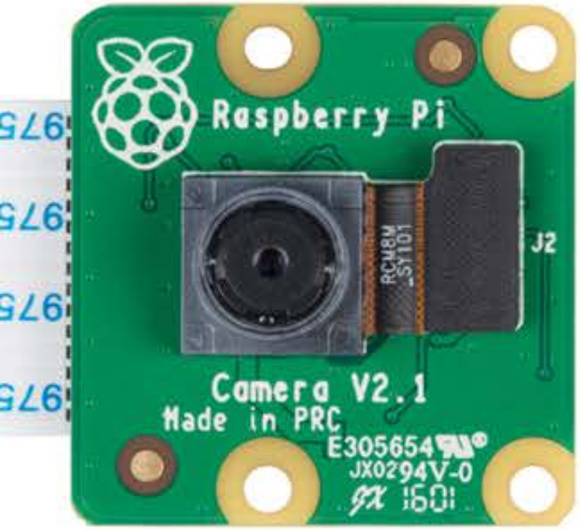
Errors like missing yarn or misaligned loops can propagate across cycles, leading to wasted material and time. Early detection of such defects also minimizes frustration and loss in creative processes.

Vision-based approach

Non-intrusive vision systems allow precise monitoring of yarn configurations, identifying stitching errors without disturbing the mechanical behavior of knitting processes.
It assures:

- non-contact,
- flexibility,
- easiness to integrate with embedded processing.

Selection of the Acquisition System



The choice of camera and interface must satisfy the following requirements:

- Compatibility with Raspberry Pi 3
- Ease of software integration
- Close-focus capability
- Cost-effectiveness
- Low power consumption

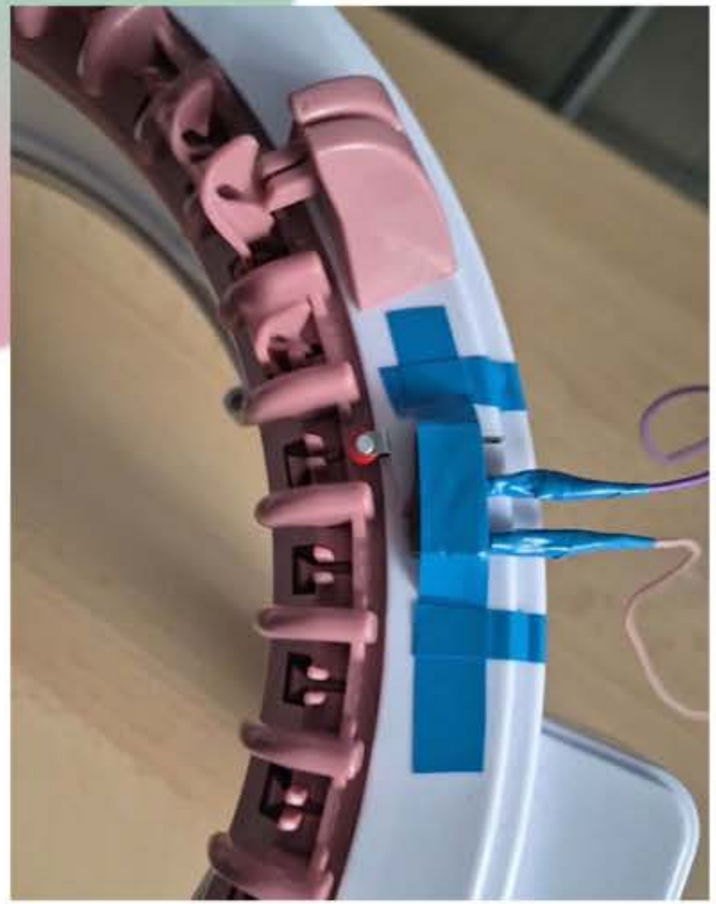
Final Choice

Pi Camera v2.1 connected via the MIPI–CSI interface emerged as the most suitable option.

It offers several advantages:

- High bandwidth and low latency
- Image quality and control
- Native compatibility with Raspberry Pi 3.

Acquisition System Design



Synchronized Image Acquisition

- A microswitch is mounted on the machine frame.
- During each cycle, a moving mechanical part activates the switch.
- The switch signal is read by the Raspberry Pi through a GPIO pin.
- When the signal is detected, the system immediately captures an image using the Pi Camera.
- This event-driven approach ensures that each image is captured at the same informative instant of the stitch formation process.

Acquisition System Design

At each trigger, the system acquires an image of the hook region and inspects the area where the stitch is expected to form.

Figure 1: Examples of correct yarn configuration

Figure 2: Examples of defective yarn configuration

Figure 3: Example image captured by the Pi Camera. The light-blue rectangle highlights the region of interest where the characteristic U-shaped yarn engagement is expected to appear

Mechanical Design of the Camera Mount

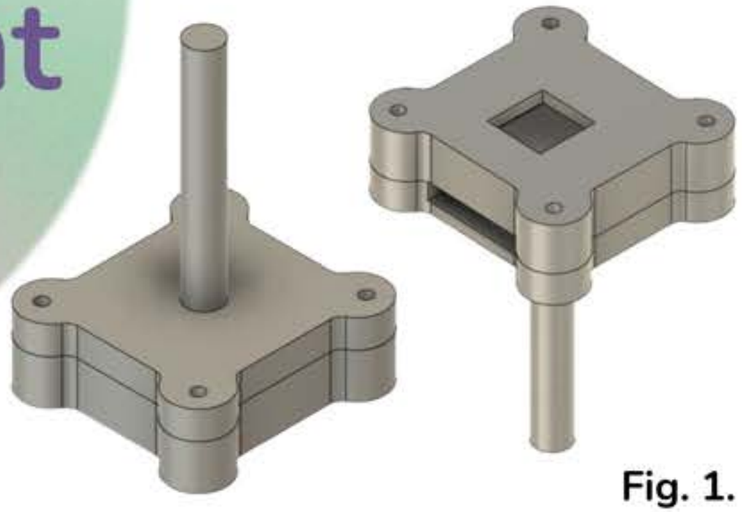
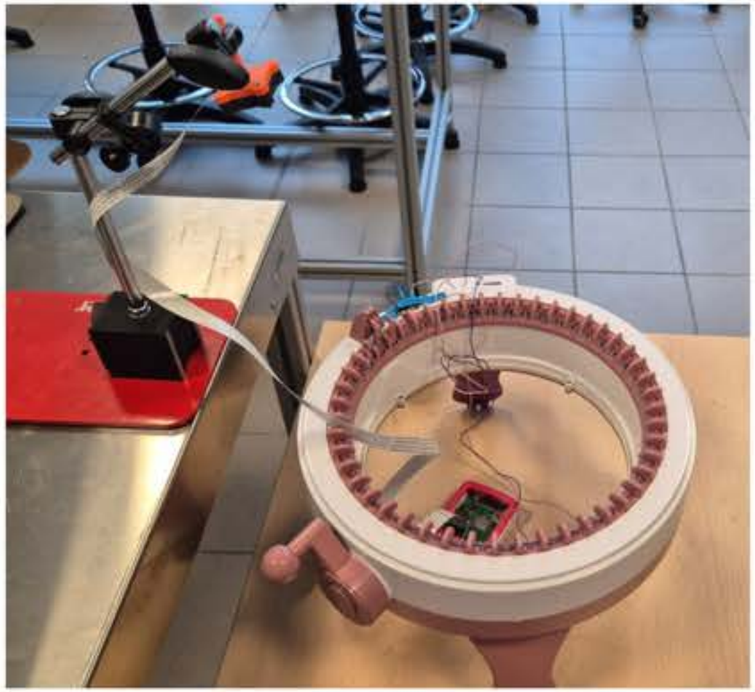


Fig. 1.

Custom Pi Camera Holder

- A custom holder for the Raspberry Pi Camera v2.1 was designed in Autodesk Fusion and fabricated using a3D printer.
- A CAD rendering of the printed holder is shown in Fig. 1.

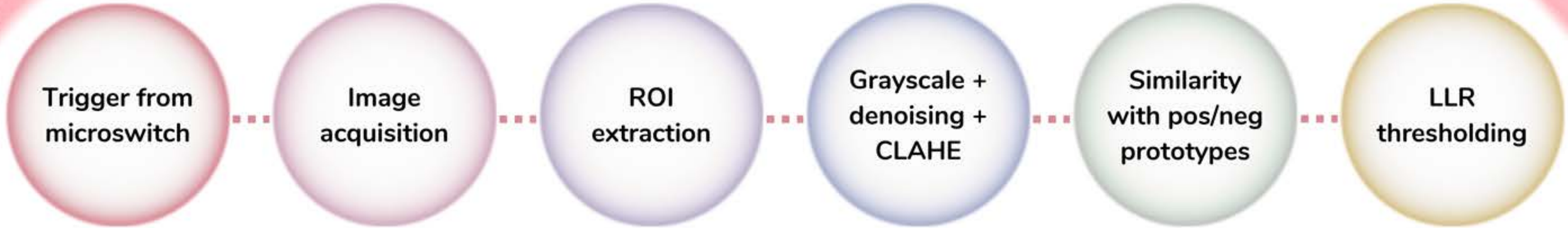
Fig. 2.



Magnetic Base

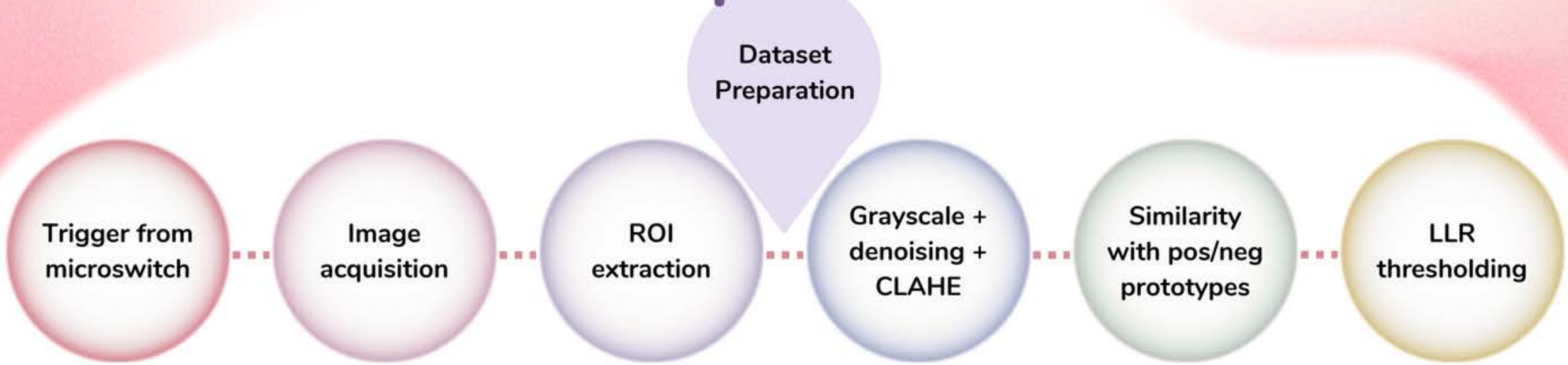
- The magnetic base provides a rigid and highly stable foundation, while allowing fine manual adjustment in height, angle, and lateral position through its articulated arm.
- The overall stiffness of the structure is not optimal yet. Improving the mechanical stability of the mounting solution is an important direction for future work.
- The assembled camera mounting system is shown in Fig. 2.

Error Detection Software Pipeline



The full pipeline runs directly on the Raspberry Pi 3 in Python and was designed to remain lightweight and deterministic

Error Detection Software Pipeline



The full pipeline runs directly on the Raspberry Pi 3 in Python and was designed to remain lightweight and deterministic

Dataset preparation

Light Blue: Examples of correct and defective yarn configuration

Purple: Examples of correct and defective yarn configuration

Fuchsia: Examples of correct and defective yarn configuration

Dataset and Color-Specific Challenges

Available yarn colors:

- light blue,
- purple,
- fuchsia.

The main challenge is not raw RGB contrast alone.

After grayscale conversion and preprocessing, different colors behave differently:

- light blue: hardest among trained colors,
- purple: intermediate behavior,
- fuchsia: surprisingly most separable.

Dataset challenges: Edge case

Out-of-Distribution Failure Case

An additional pale pink yarn was tested as an edge case.

It was almost indistinguishable from the machine background in grayscale, and the handcrafted pipeline failed to reliably separate the two classes.

This shows that:

- The method works within normal operating conditions,
- Extremely low-contrast scenarios remain out of scope,
- More advanced learned features may be needed in the future.

Fig. 1: Available yarn colors

Fig. 2: Pale Pink: Edge case, example of correct and defective configuration

Dataset Structure

Dataset was partitioned into three non-overlapping subsets:

- Training set: 39 positive + 27 negative images,
- Validation set: 208 images,
- Test set: 771 images.

Training set

- Used to build the reference prototypes for the two classes.
- Positive and negative images are aggregated to compute the median templates used by the classifier.

Validation set

- Used to select the operating threshold of the LLR detector.
- Allows tuning the decision rule without influencing the final evaluation.

Test set

- Used only for the final performance evaluation.
- Provides an unbiased estimate of the system accuracy and detection capability.

Discriminative Mask Generation from Training Set

Grayscale Conversion

The RGB image is converted to grayscale in order to **remove color information** and retain only the intensity structure of the yarn and hook region.

Gaussian Filtering

A small Gaussian filter is applied to reduce **high-frequency noise** while preserving the main structural features of the yarn and hook.

CLACHE

CLAHE is applied to **enhance local contrast**, allowing subtle structures such as yarn texture and shadows to remain visible even under non-uniform illumination.

Fisher-like Separability Score

A Fisher-like separability score is computed for each pixel to measure how well it **distinguishes positive** and **negative** samples.

$$S(i,j) = \frac{|\mu_{pos}(i,j) - \mu_{neg}(i,j)|}{\sigma_{pos}(i,j) + \sigma_{neg}(i,j) + \varepsilon}$$

Otsu Thresholding

Otsu thresholding is used to automatically convert the soft separability map into a **binary mask** that highlights the most discriminative pixels.

Discriminative Mask Generation from Training Set

Fisher-like Separability Score

Fig. 1: Soft Mask

Fig. 2: Hard Mask

Otsu Thresholding

$$S(i,j) = \frac{|\mu_{pos}(i,j) - \mu_{neg}(i,j)|}{\sigma_{pos}(i,j) + \sigma_{neg}(i,j) + \varepsilon}$$

$\mu_{pos}(i,j)$, $\mu_{neg}(i,j)$ } mean intensities of the positive and negative training images

$\sigma_{pos}(i,j)$, $\sigma_{neg}(i,j)$ } corresponding standard deviations

The resulting soft score map was then converted into an 8-bit image and thresholded using Otsu's method, which automatically selects the threshold that best separates low-score and high-score pixels.

This produces a binary hard mask that retains only the most discriminative regions of the hook area. However, this representation discards part of the local texture information, and therefore it was not selected for the final classifier, which instead uses the soft mask

Masked Similarity Computation (NCC and LLR)

Discriminative Mask Generation from the Training Set

After generating the soft mask, the classifier compares the input crop with two reference templates:

- a positive template representing correct yarn engagement
- a negative template representing missing-yarn defects

The positive and negative templates are obtained by averaging the training images of the two classes.

The comparison is performed using a masked normalized cross-correlation (NCC), where the soft mask acts as a weighting function that emphasizes the most informative pixels.

$$s_{\text{pos}} = \text{NCC}_w(I, \mu_{\text{pos}}) \quad s_{\text{neg}} = \text{NCC}_w(I, \mu_{\text{neg}})$$

LLR-Based Decision Rule

The two scores are then combined into a log-likelihood-ratio-like decision variable: $LLR = s_{\text{pos}} - s_{\text{neg}}$

The LLR compares how much the input image resembles the positive template versus the negative one. This allows the decision to be based on the relative evidence between the two hypotheses rather than on a single similarity score.

Validation Set

Threshold Selection Using the Validation Set

- The decision threshold for the LLR classifier was selected using the validation set. Different threshold values were evaluated in order to study the trade-off between false positives and false negatives.
- In the context of the crochet machine, false positives are particularly undesirable, since they would trigger unnecessary machine stops and interrupt the production process.
- For this reason, the threshold was chosen to achieve a false positive rate close to zero, even at the cost of a moderate increase in false negatives.
- The selected threshold value was therefore: $\mathrm{LLR}_{\mathrm{thr}} = -0.035$

Evaluation on the Test Set

Confusion Matrix on the Test Set

After selecting the classifier threshold using the validation set, the final performance was evaluated on an independent test set using a confusion matrix.

The test set was never used during the training or threshold selection phases, ensuring an unbiased estimate of the system performance

Table 1. Confusion matrix on the test set (771 images)

	Predicted H_0	**Predicted H_1**
True H_0 (no defect)	TN = 633	FP = 0
True H_1 (defect)	FN = 20	TP = 118

- Overall Accuracy: 97.4%
- TPR: 85.5%
- FPR: 0%

Fuschia Confusion Matrix

- Overall Accuracy: 99.2%
- TPR: 94.1%
- FPR: 0%

	Predicted H_0	Predicted H_1
True H_0	TN = 322	FP = 0
True H_1	FN = 3	TP = 48

Purple Confusion Matrix

- Overall Accuracy: 96.6%
- TPR: 82.2%
- FPR: 0%

	Predicted H_0	Predicted H_1
True H_0	TN = 193	FP = 0
True H_1	FN = 8	TP = 37

Light Blue Confusion Matrix

- Overall Accuracy: 94.4%
- TPR: 78.6%
- FPR: 0%

	Predicted H_0	Predicted H_1
True H_0	TN = 118	FP = 0
True H_1	FN = 9	TP = 33

Real-Time Operation

Synchronization and Timing constraints

The inspection process is synchronized with the crochet machine cycle through a microswitch. The machine activates the switch approximately every 0.42 seconds, corresponding to:

$$f_{\text{inspect}} = \frac{1}{0.42} \approx 2.4 \,\text{inspections per second}$$

For each trigger:

- an image is captured,
- the hook region is cropped,
- preprocessing and classification are performed.

The computational cost of the classifier is negligible compared to the machine cycle time. This suggests that multiple images could be acquired for each stitch and combined to improve the overall detection accuracy.

Limitations and Future Work

Current limitations	Possible improvements
Some missed detections could still be present in challenging visual conditions. Performance degrades in very low-contrast conditions. The system currently relies on a single image for each stitch inspection	Acquire multiple images per stitch and combine the results through a multi-frame decision strategy. Improve the mechanical stiffness of the camera mount to reduce small misalignments. Explore lightweight learning-based feature extraction methods to improve robustness.

Thank you for your attention